

课程简介

- 课程要求
- 课程教材
- 课程参考PPT
- 课程考核标准
- 课程目标

课程要求

- 课程预先要求知识：
 - 概率论
 - 线性代数
 - 多元微积分
 - Python语言(Matlab)
- 不要求图像处理，计算机视觉和计算机图形学知识

教材

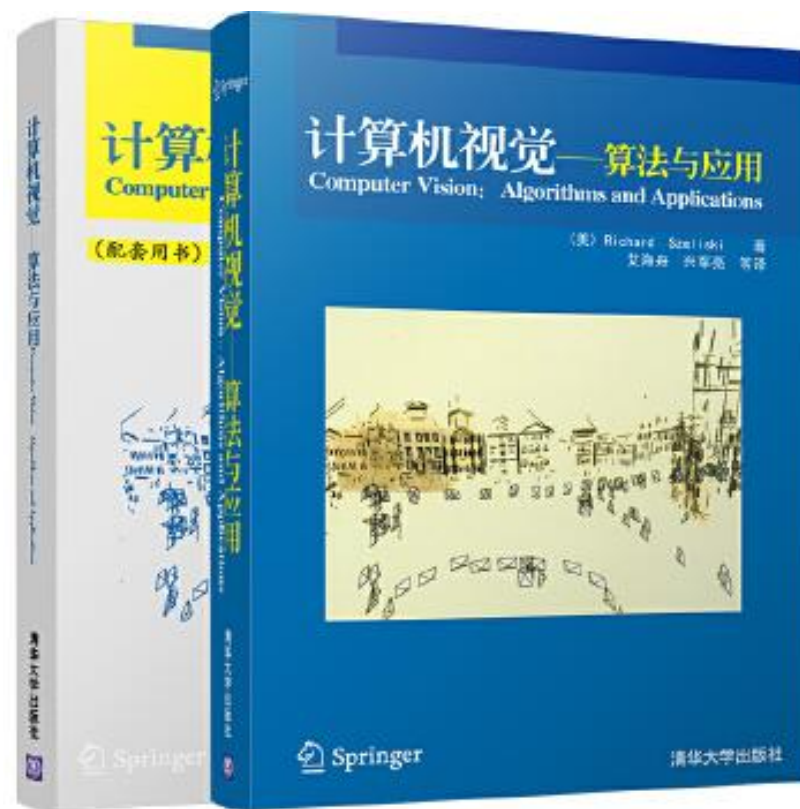
- 1. Richard Szeliski Computer Vision: Algorithms and Applications 2nd Edition.
- 2. E. R. Davies Computer Vision: Principles, Algorithms, Applications, Learning(5th Edition)
- 3.Simon J.D. Prince Computer Vision: Models, Learning, and Inference
- 4. Forsyth and Ponce, Computer Vision: A Modern Approach 2nd Edition.

教材一

- Rick Szeliski 计算机视觉：算法与应用 第二版

艾海舟等译

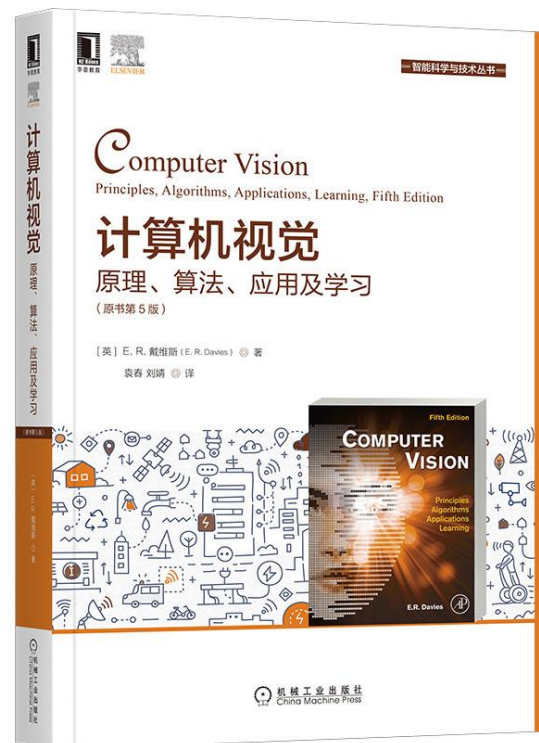
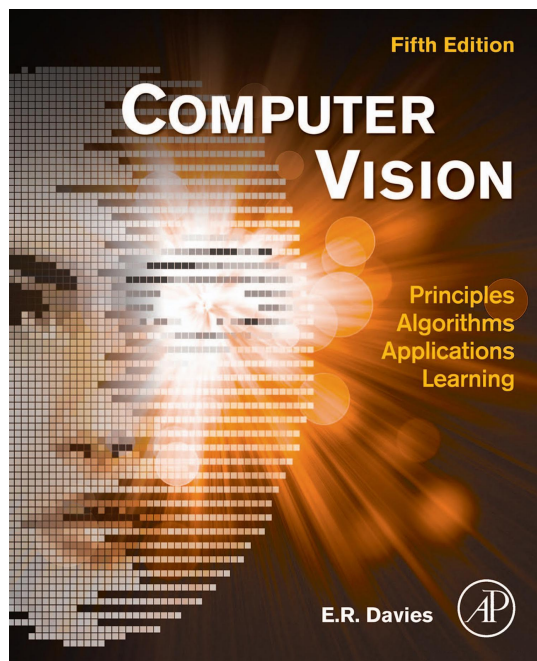
清华大学出版社



教材二

- E. R. Davies 计算机视觉：原理、算法、应用及学习（第五版）

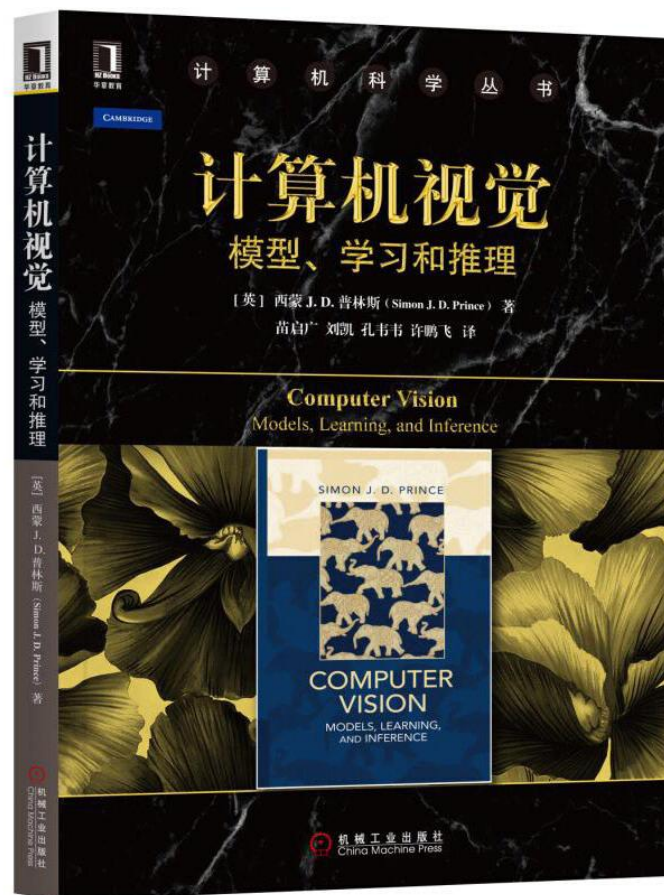
机械工业出版社



参考书一

- 西蒙 J.D. 普林斯 计算机视觉：模型、学习和推理

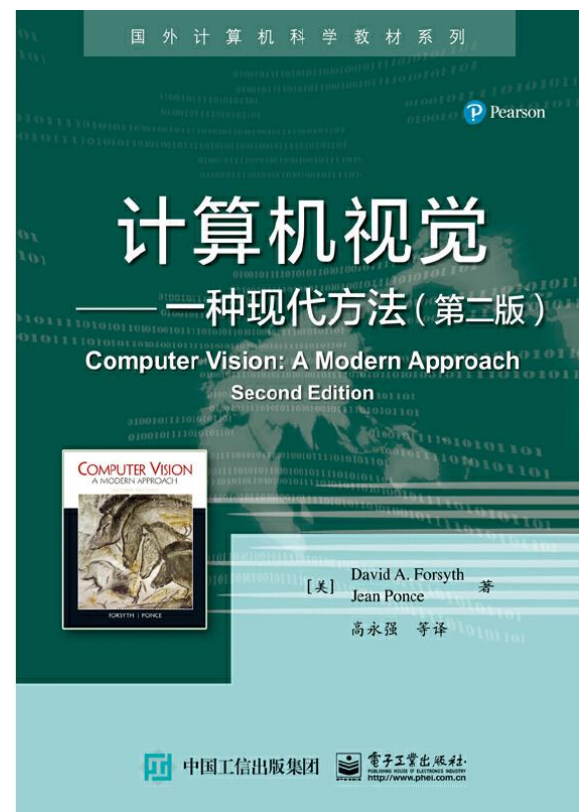
机械工业出版社



参考书二

- Forsyth and Ponce**计算机视觉——一种现代方法**

电子工业出版社



讲义PPT

- 主要参考 PPT :

1.Prof. Trevor Darrell's C280 , Computer Vision.

2.Rick Szeliski's CSE 576;

<https://courses.cs.washington.edu/courses/cse576/20sp/calendar/>

3.Justin Johnson's EECS 498-007 , Deep Learning for Computer Vision.

<https://web.eecs.umich.edu/~justincj/teaching/eecs498/FA2020/schedule.html>

参考 PPT



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Calendar — CSE 576 Spring 2020

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courses.cs.washington.edu/courses/cse576/20sp/calendar/

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CSE 576

About

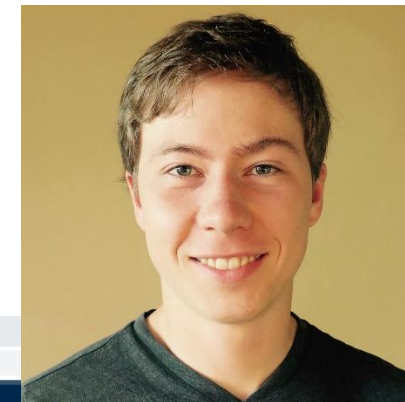
Calendar

Calendar

- Slides are linked in the table below.
- Lecture recordings are available through [Canvas](#).
- Assignments are due by 3pm (before the lecture) on the due date.

Date	Topic	Lecturer	Slides	Reading	Homework
March 31	Introduction	Steve Seitz	Google Slides	Szeliski, Chapter 1	
April 2	Human Vision, Color Spaces and Transforms	Steve Seitz	Google Slides	Brown, M. S. (2019). ICCV 2019 tutorial on understanding color and the in-camera image processing pipeline for computer vision.	HW1 assigned
April 7	Image coordinates, resizing	Steve Seitz	Google Slides	Szeliski, Chapter 2.1, 3.6	
April 9	Filters and convolutions	Steve Seitz	Google Slides	Szeliski, Chapter 3	HW1 due, HW2 assigned
April 14	Interpolation and Optimization	Rick Szeliski	pdf , pptx	Szeliski, Chapter 4	
April 16	Machine Learning	Rick Szeliski	pdf , pptx	Szeliski, Chapter 5.1-5.2	
April 21	Neural Networks	Rick Szeliski	pdf , pptx	Szeliski, Chapter 5.3	

参考 PPT



Schedule | EECS 498-007 / 598-005

web.eecs.umich.edu/~justincj/teaching/eecs498/FA2020/schedule.html

EECS 498-007 / 598-005
Deep Learning for Computer Vision
Fall 2020

Schedule

Lectures will be Mondays and Wednesdays 1:30 - 3pm on Zoom. Attendance is not required. Recordings will be posted after each lecture in case you are unable to attend the scheduled time.

Some lectures have reading drawn from the course notes of [Stanford CS 231n](#), written by [Andrej Karpathy](#).

Some lectures have optional reading from the book *Deep Learning* by Ian Goodfellow, Yoshua Bengio, and Aaron Courville (GBC for short). The entire text of the book is available [for free online](#) so you don't need to buy a copy.

Event	Date	Description	Course Materials
Lecture 1	Monday August 31	Course Introduction Computer vision overview Historical context Course logistics	[slides] [FA2020 video (UMich only)] [FA2019 video (public)] [Python tutorial] [GBC Sec 1.2] [GBC Sec 6.6]
Lecture 2	Wednesday September 2	Image Classification Data-driven approach K-Nearest Neighbor Hyperparameters Cross-validation	[slides] [FA2020 video (UMich only)] [FA2019 video (public)] [231n Image Classification]
	Monday September 7	No class Labor Day	
Lecture 3	Wednesday September 9	Linear Classifiers Algebraic / Visual / Geometric viewpoints Softmax / SVM classifiers	[slides] [FA2020 video (UMich only)] [FA2019 video (public)] [231n Linear Classification]
A1 Due	Friday	Assignment 1 Due PyTorch warmup	[Assignment 1]

课程考核标准

- 课程考核

- 考勤 5%
- 平时作业+实验 25%
- 期末考试 70%

考勤(5%)

- 课堂出勤
 - 形式
 - 随机点名
- 成绩评定
 - 一次未到：-20

作业 (10%-15%)

- 课堂作业

- 形式

- 每两周一次 (共3次) , 完成期限一周

- 内容

- 课程重点内容练习, 使用matlab/python编程实现
 - 华为昇腾平台实验

- 成绩评定 (共3次)

- 未完成0分
 - 完成不好酌情减分

实验（10%）

- 两次实验

- 形式

- 上机实验（共2次）

- 内容

- 华为AI平台实验

- 1. 物体识别 分类模型
 - 2. 图像分割 分割模型

4课时

4课时

- 成绩评定

- 实验报告评分
 - 完成不好酌情减分

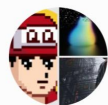
期末（70%-75%）

- 课堂内容考核
 - 形式
 - 笔试
 - 内容
 - 基本概念
 - 核心知识点
 - 课堂作业或实验升级
- 成绩评定
 - 卷面分

课程目标

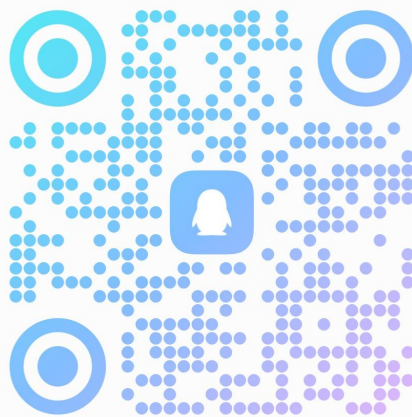
- 1. 图像形成的基本原理
- 2. 图像分析基础
 - 卷积与图像金字塔
 - 图像局部特征分析
- 3. 基于深度学习的图像分析
 - 深度学习与图像分类
 - 图像分割
 - 目标检测与识别
 - 图像生成
- 4. 视觉分析
 - 图像全景拼接与配准
 - 双目视觉原理

课程QQ群



2025秋本计...

群号: 650369780



扫一扫二维码，入群聊

